Experiment #5

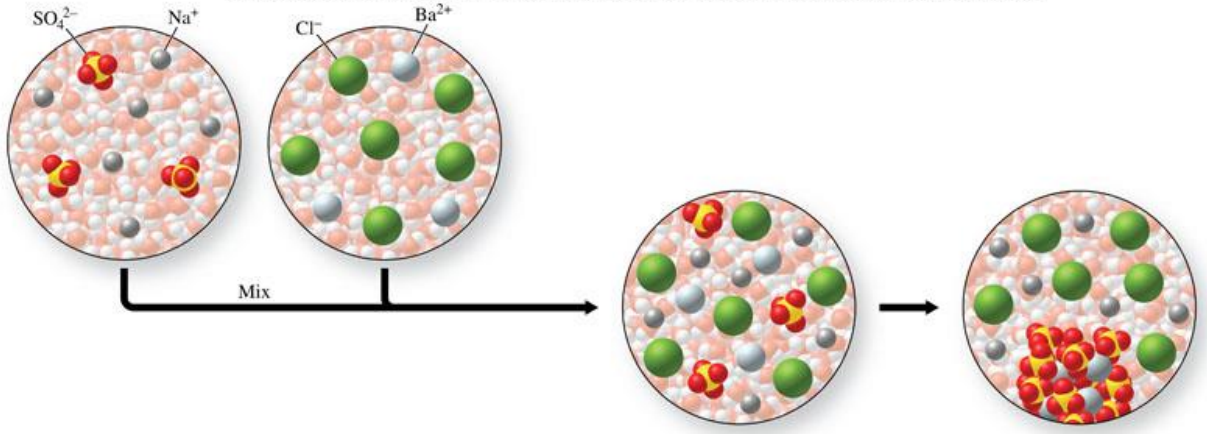
Qualitative Analysis of Cations in Solution

Precipitation Reaction

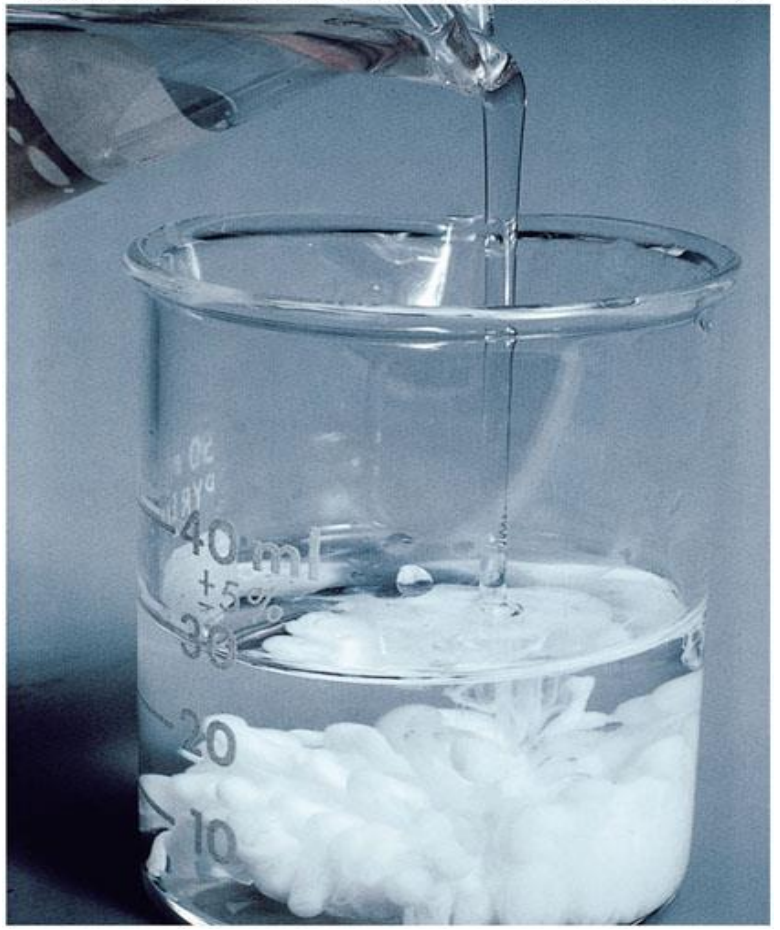
In a precipitation reaction, an insoluble solid called a precipitate is formed.

$$BaCl_2(aq) + Na_2SO_4(aq) \rightarrow BaSO_4(s) + 2NaCl(aq)$$

Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.



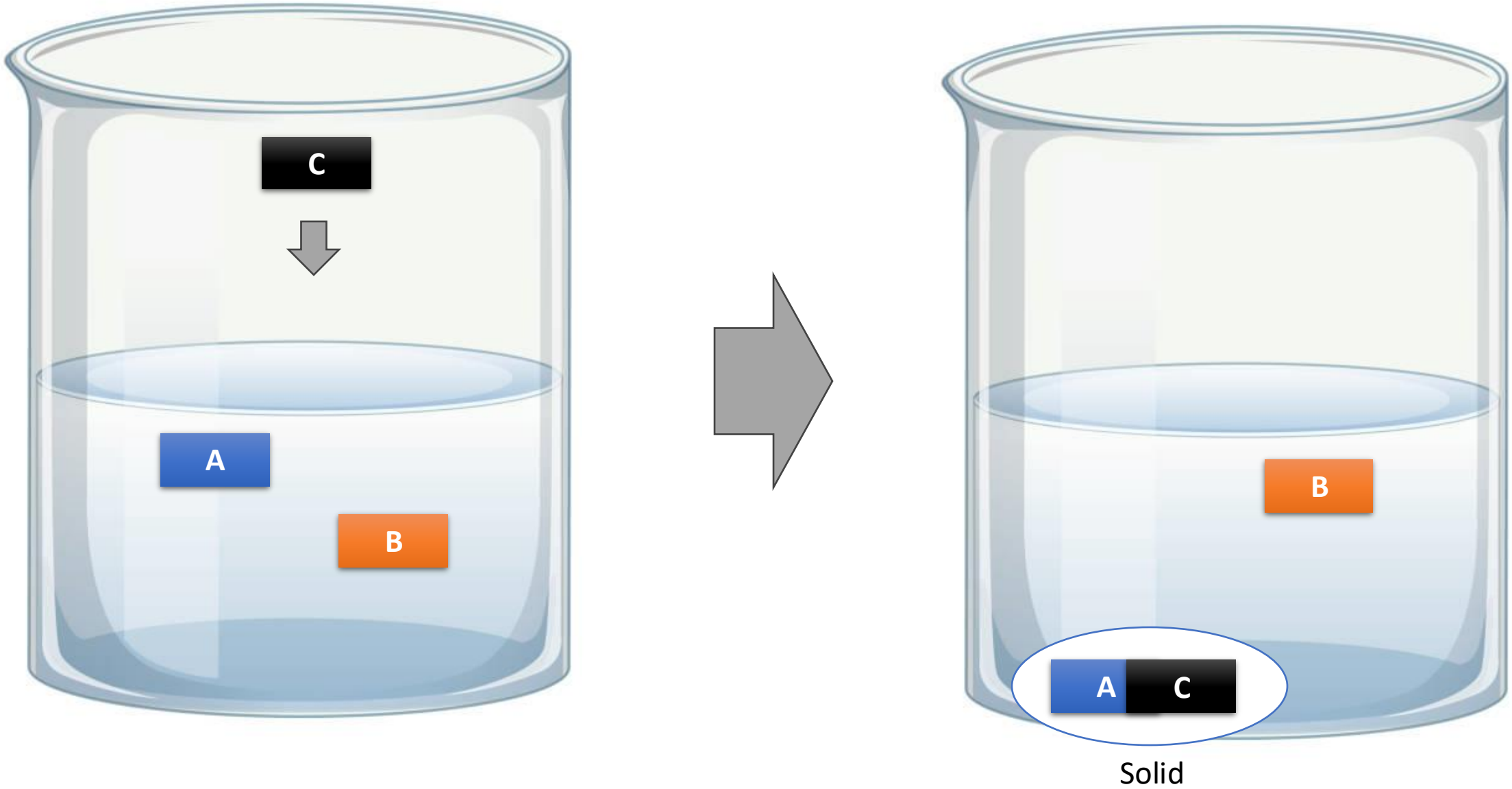
Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.



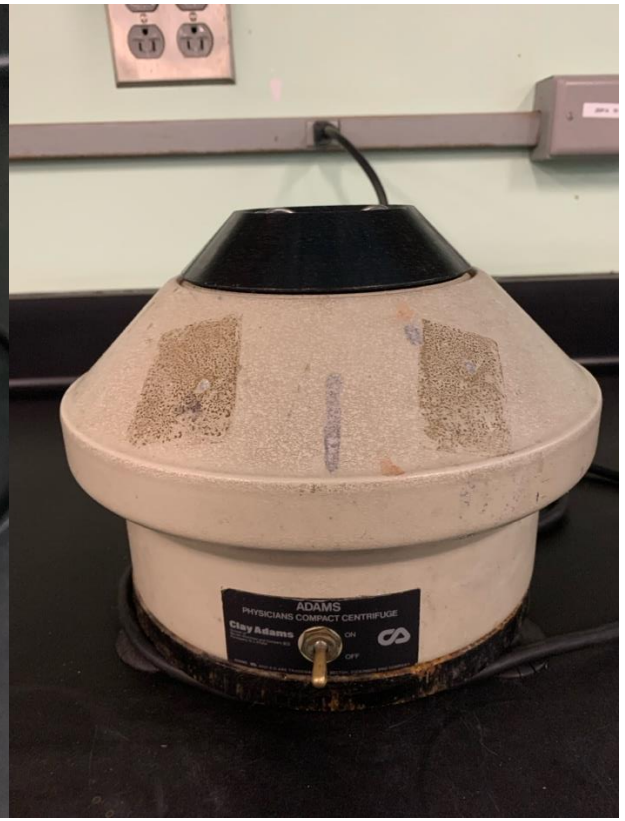
$$BaCl_2(aq) + Na_2SO_4(aq) \rightarrow BaSO_4(s) + 2NaCl(aq)$$

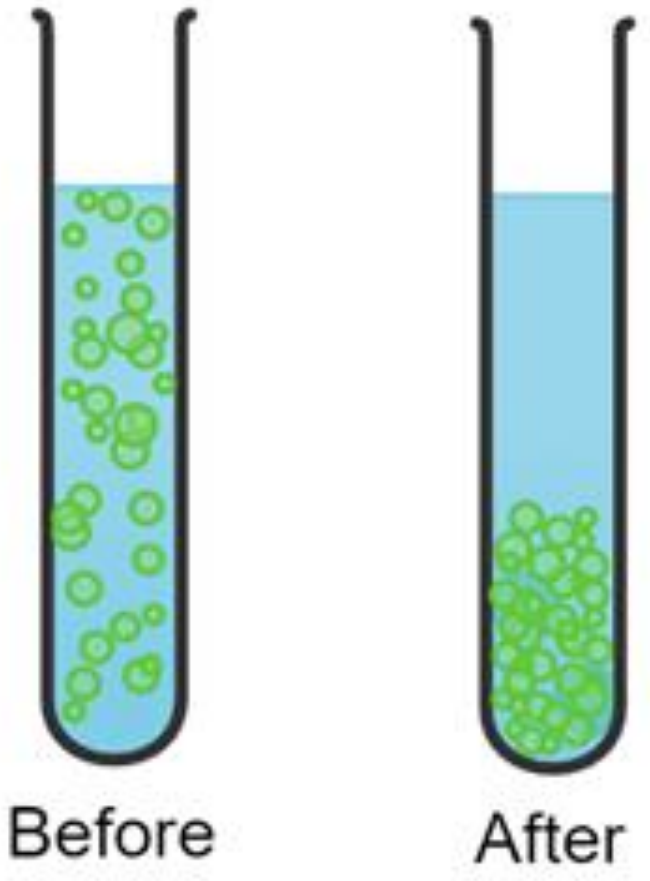
TABLE 5.4 | Rules Used to Predict the Solubility of Ionic Compounds

Ions	Rule
Na^+, K^+, NH_4^+ (and other alkali metal ions)	Most compounds of alkali metal and ammonium ions are soluble.
NO_3^-, $CH_3CO_2^-$	All nitrates and acetates are soluble.
SO_4^{2-}	Most sulfates are soluble. Exceptions are $BaSO_4$, $SrSO_4$, $PbSO_4$, $CaSO_4$, Hg_2SO_4, and Ag_2SO_4.
Cl^-, Br^-, I^-	Most chlorides, bromides, and iodides are soluble. Exceptions are AgX, Hg_2X_2, PbX_2, and HgI_2. (X = Cl, Br, or I.)
Ag^+	Silver compounds, except $AgNO_3$ and $AgClO_4$, are insoluble. $AgCH_3CO_2$ is slightly soluble.
O^{2-}, OH^-	Oxides and hydroxides are insoluble. Exceptions are alkali metal hydroxides, $Ba(OH)_2$, $Sr(OH)_2$, and $Ca(OH)_2$ (somewhat soluble).
S^{2-}	Sulfides are insoluble. Exceptions are compounds of Na^+, K^+, NH_4^+ and the alkaline earth metal ions.
CrO_4^{2-}	Most chromates are insoluble. Exceptions are compounds of Na^+, K^+, NH_4^+, Mg^{2+}, Ca^{2+}, Al^{3+}, and Ni^{2+}.
CO_3^{2-}, PO_4^{3-}, SO_3^{2-}, SiO_3^{2-}	Most carbonates, phosphates, sulfites, and silicates are insoluble. Exceptions are compounds of Na^+, K^+, and NH_4^+.

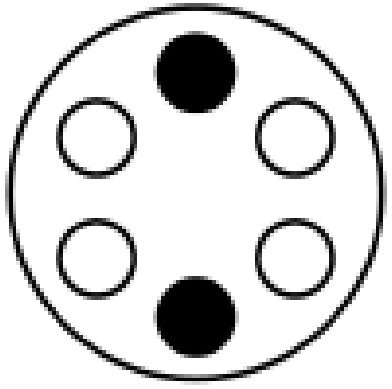

C
A
B
B
A
C
Solid

Centrifuge

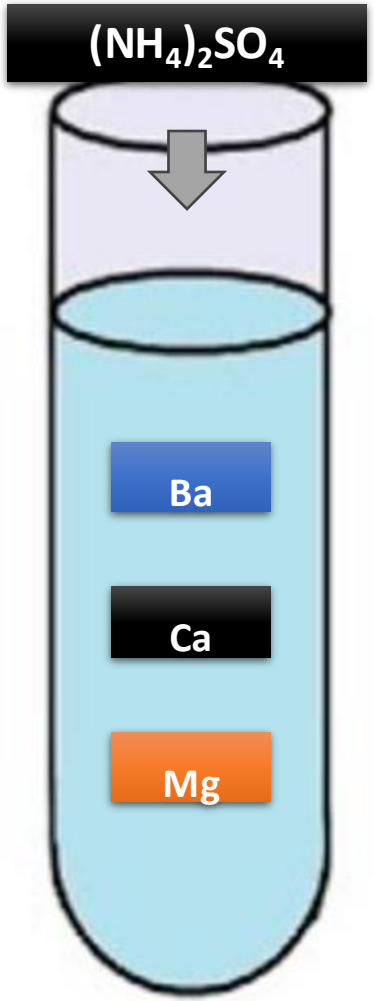




https://www.youtube.com/shorts/7SOz0HyaRdw?feature=share

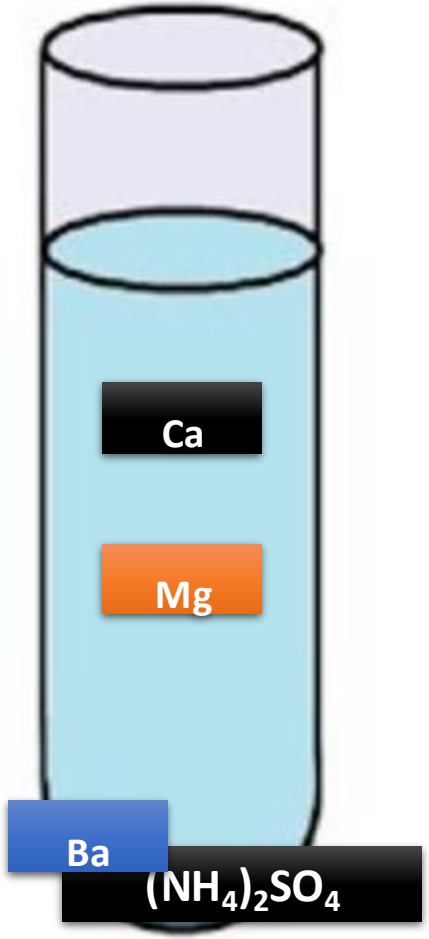


Centrifuge loading pattern for two test tubes (black circles) or four test tubes (white circles).



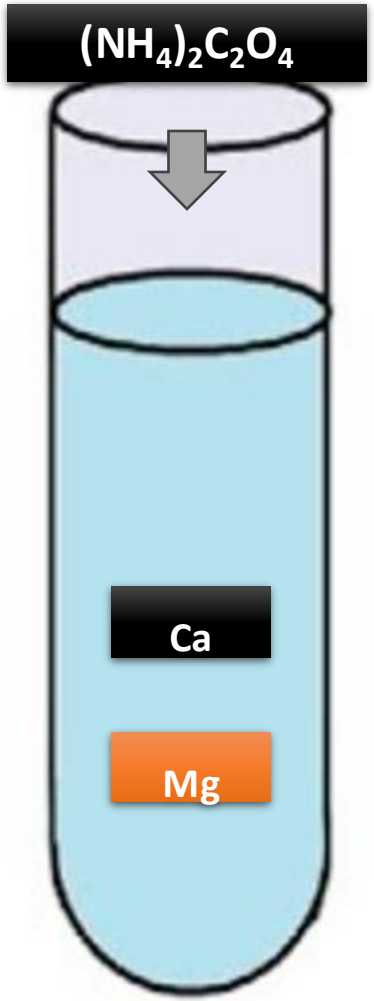
Test Tube #1

1) Mix
2) Centrifuge



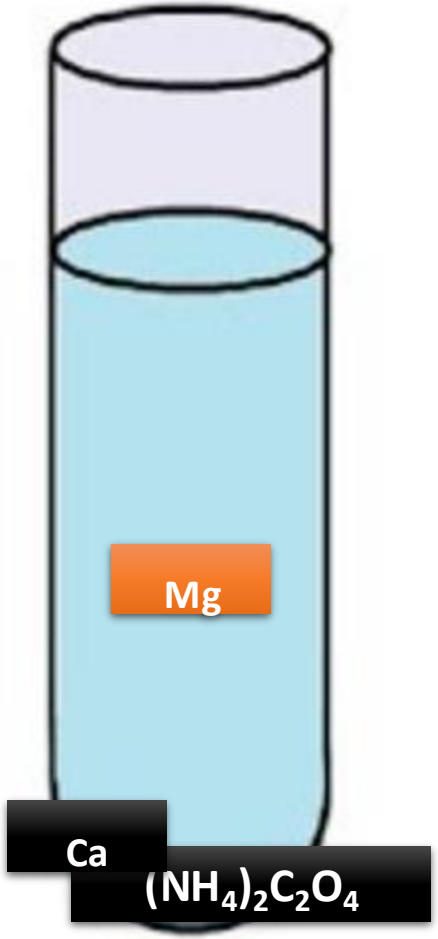
Pour the liquid to test tube #2

1) Add 10-15 drops of HCl to solid
2) Flame test (yellow green)



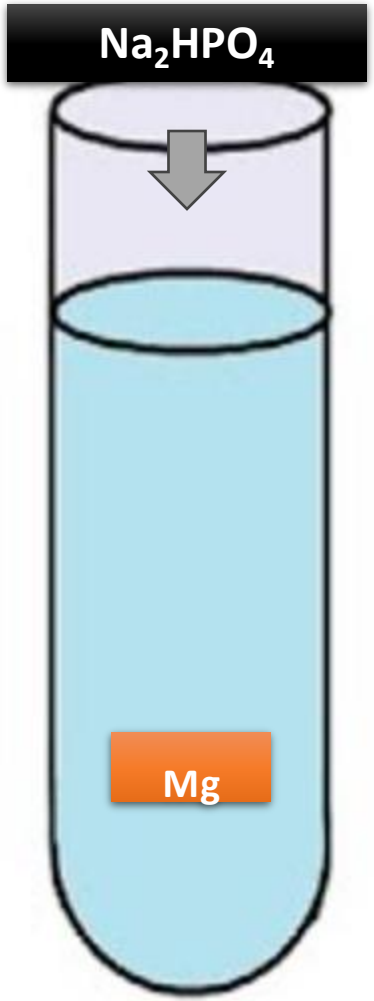
Test Tube #2

1) Mix
2) Centrifuge



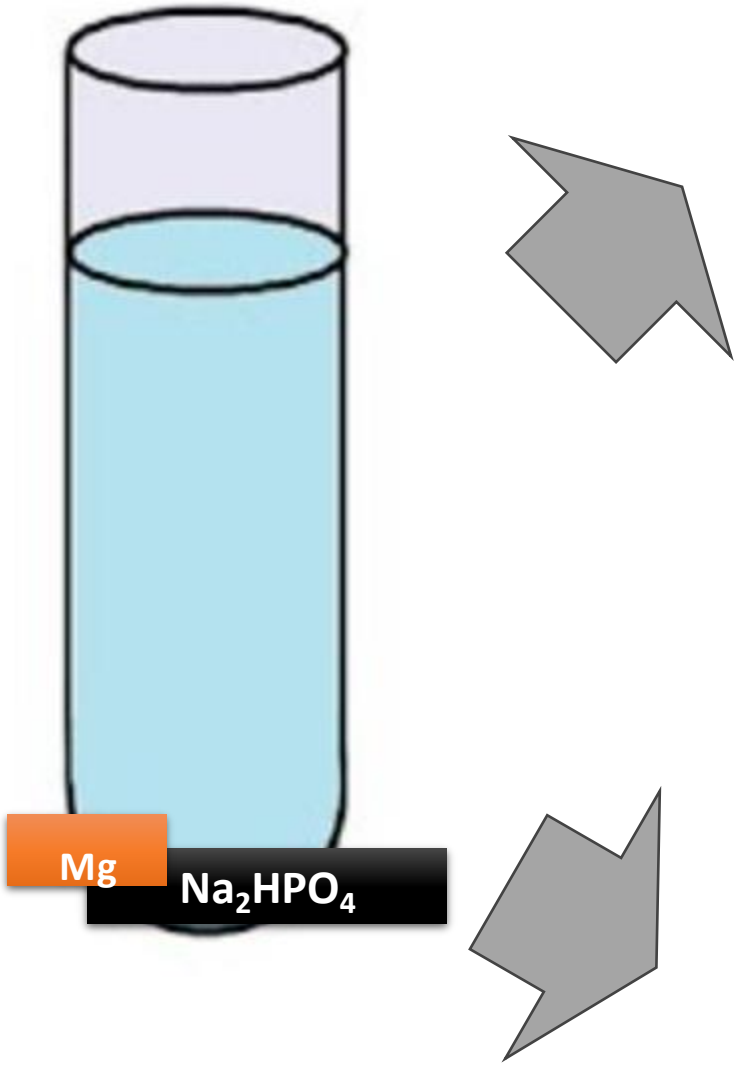
Pour the liquid to test tube #3

1) Add 5 drops of HCl to solid
2) Flame test (orange red)



Test Tube #2

1) Mix
2) Centrifuge

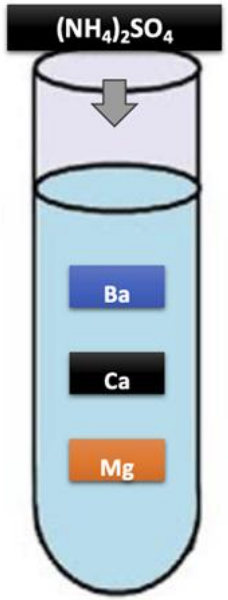


Waste Container Beaker

Blue lake test

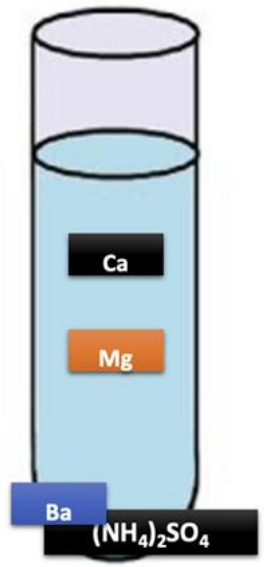
Blue lake test

1) **add Mg indicator (8 drops)**
2) **Add NaOH dropwise until red litmus paper turns blue**
3) **Centrifuge**
4) **Blue gel = Presence of Mg**



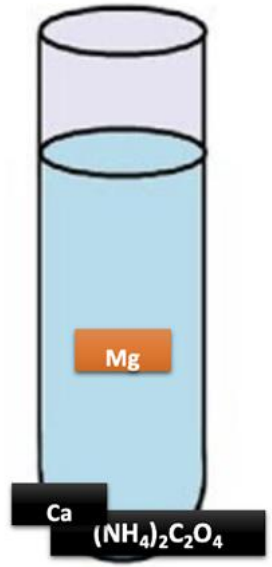
Test Tube #1

1) Mix
2) Centrifuge



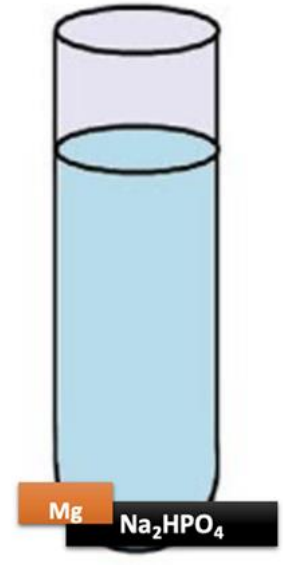
1) Add 10-15 drops of HCl to solid
2) Flame test (yellow green)

1) Mix
2) Centrifuge



1) Add 5 drops of HCl to solid
2) Flame test (orange red)

1) Mix
2) Centrifuge



Blue lake test

1) add Mg indicator (8 drops)
2) Add NaOH dropwise until red litmus paper turns blue
3) Centrifuge
4) Blue gel = Presence of Mg